ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

ELECTRICAL CIRCUITS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 1st year Electrical Circuits paper 1 · practice a.

Written by Academic Central for practice, following first-year Electrical Circuits syllabuses at South African universities (DC and AC circuit analysis for engineering students). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct method (for example mesh analysis instead of nodal analysis) and answers that differ only through rounding.

QUESTION 1

1.1 Current: the rate of flow of charge, I = Q/t, unit ampere (A) ✓ (3)
Potential difference: the energy per unit charge between two points, V = W/Q, unit volt (V) ✓
Resistance: the opposition to current, R = V/I, unit ohm (Ω) ✓

1.2 $R = \rho L / A$ ✓ $= 1.72 \times 10^{-8} \times 100 / 1.5 \times 10^{-6}$ ✓ $= 1.15\ \Omega$ ✓ (3)

1.3 $V = 12 \times 8 / (4 + 8)$ ✓ $= 8$ V ✓ (2)

1.4.1 $Q = It = 2.5 \times 3 \times 3\,600$ ✓ $= 27\,000$ C ✓ (2)

1.4.2 $W = VQ = 12 \times 27\,000 = 324\,000$ J ✓ $= 0.09$ kWh ✓ (2)

1.5.1 1 kΩ ($10 \times 10^2\ \Omega$), ±5% ✓ (1)

1.5.2 $P = V^2/R = 12^2 / 1\,000 = 0.144$ W ✓ (2)
0.144 W is less than 0.25 W, so the resistor is suitable ✓

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QUESTION 2

2.1 $R2 \parallel R3 = (12 \times 6) / (12 + 6)$ ✓ $= 4\ \Omega$ ✓ (4)
$RT = 4 + 4 + 2$ ✓ $= 10\ \Omega$ ✓

2.2 $I = V / RT = 24 / 10$ ✓ $= 2.4$ A ✓ (2)

2.3 $V_{R1} = 2.4 \times 4 = 9.6$ V ✓ (4)
$V_{parallel} = 2.4 \times 4 = 9.6$ V ✓
$V_{R4} = 2.4 \times 2 = 4.8$ V ✓
(Check: 9.6 + 9.6 + 4.8 = 24 V) ✓

2.4 $I_2 = 9.6 / 12 = 0.8$ A ✓ $I_3 = 9.6 / 6 = 1.6$ A ✓ (3)
(Check: 0.8 + 1.6 = 2.4 A, KCL) ✓

2.5 $P = VI = 24 \times 2.4$ ✓ $= 57.6$ W ✓ (4)
$P_{R3} = I_3^2 R3 = 1.6^2 \times 6$ ✓ $= 15.36$ W ✓

2.6 Sum of the voltage drops: $9.6 + (0.8 \times 12) + 4.8$ ✓ $= 9.6 + 9.6 + 4.8 = 24$ V ✓ (3)
This equals the supply EMF, so the sum of the voltages around the loop is zero ✓

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QUESTION 3

3.1.1 Gain $= -Rf / Rin = -100 / 10 = -10$ ✓ (2)
$V_{out} = -10 \times 0.5 = -5$ V ✓

3.1.2 Any TWO: infinite input resistance (no current into the inputs); zero output resistance; infinite open-loop gain; the two inputs are at the same voltage when negative feedback is used; infinite bandwidth ✓✓ (2)

3.2.1 Let V be the voltage of the top node of R3 (bottom wire = 0 V) ✓ (6)
(V − 10)/2 + (V − 5)/1 + V/4 = 0 ✓✓
2V − 20 + 4V − 20 + V = 0, so 7V = 40 ✓
V = 5.71 V ✓ I_3 = 5.714 / 4 = 1.43 A ✓

3.2.2 I_1 = (10 − 5.714) / 2 = 2.14 A, delivered by E1 (out of its positive terminal) ✓ (3)
I_2 = (5 − 5.714) / 1 = −0.71 A ✓
The minus sign means 0.71 A flows INTO the positive terminal of E2, so E2 is being charged ✓

3.3.1 With R3 removed, loop current = (10 − 5) / (2 + 1) = 1.667 A ✓ (3)
Vth = 10 − 2 × 1.667 ✓ = 6.67 V ✓

3.3.2 Short-circuit both sources: Rth = 2 ‖ 1 = (2 × 1) / 3 ✓ = 0.67 Ω ✓ (2)

3.3.3 I = Vth / (Rth + R3) = 6.67 / 4.67 ✓ = 1.43 A, the same as in 3.2.1 ✓ (2)

3.3.4 I_N = Vth / Rth = 6.67 / 0.667 = 10 A (the same as the short-circuit current 10/2 + 5/1) ✓ (2)
R_N = Rth = 0.67 Ω, connected in parallel with the current source ✓

3.4 Maximum power when R_L = Rth = 0.67 Ω ✓ (3)
P_max = V_{th}^2 / (4Rth) = 6.67^2 / (4 × 0.667) ✓ = 16.67 W ✓

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QUESTION 4

4.1 Q = CV = 470 × 10^{-6} × 12 ✓ = 5.64 mC ✓ (4)
W = ½CV^2 = ½ × 470 × 10^{-6} × 12^2 ✓ = 33.84 mJ ✓

4.2.1 1/C = 1/10 + 1/20 + 1/30 ✓ C = 5.45 µF ✓ (2)

4.2.2 C = 10 + 20 + 30 = 60 µF ✓ (1)

4.3.1 τ = RC = 10 000 × 100 × 10^{-6} = 1 s ✓ (1)

4.3.2 v_C = V(1 − e^(−t/τ)) ✓ = 12(1 − e^{-2}) ✓ = 10.38 V ✓ (3)

4.3.3 After about 5τ = 5 s ✓ (1)

4.3.4 Voltage rises quickly at first and then levels off towards 12 V (rising exponential) ✓ (2)
Current starts at its highest value (12 V / 10 kΩ = 1.2 mA) and falls towards zero (falling exponential) ✓

4.4 e = L (di/dt) = 0.5 × (2 / 0.01) ✓ = 100 V ✓ (4)
W = ½LI^2 = ½ × 0.5 × 2^2 ✓ = 1 J ✓

4.5 When the transistor switches off, the collapsing magnetic field of the coil induces a large reverse voltage (back EMF) ✓ (2)
The diode gives this current a safe path, so the voltage spike does not damage the transistor ✓

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QUESTION 5

5.1 Vpeak = 325 V ✓ (4)
Vrms = 325 / √2 = 229.81 V ✓
f = 314.16 / 2π = 50 Hz ✓
T = 1 / 50 = 0.02 s (20 ms) ✓

5.2.1 $X_L = 2\pi fL = 2\pi \times 50 \times 0.1 = 31.42\ \Omega$ ✓ (2)
$X_C = 1 / (2\pi fC) = 1 / (2\pi \times 50 \times 50 \times 10^{-6}) = 63.66\ \Omega$ ✓

5.2.2 $Z = \sqrt{(R^2 + (X_L - X_C)^2)}$ ✓ $= \sqrt{(40^2 + (31.42 - 63.66)^2)}$ ✓ $= 51.38\ \Omega$ ✓ (3)

5.2.3 $I = V / Z = 230 / 51.38 = 4.48$ A ✓ (1)

5.2.4 $\varphi = \tan^{-1}((X_L - X_C) / R) = \tan^{-1}(-32.24 / 40) = -38.87°$ ✓ (3)
Power factor = $\cos \varphi = R / Z = 0.78$ ✓
$X_C > X_L$, so the circuit is capacitive and the current LEADS the voltage ✓

5.2.5 $P = I^2R = 4.476^2 \times 40$ ✓ $= 801.41$ W ✓ (or $P = VI \cos \varphi$) (2)

5.2.6 $f_0 = 1 / (2\pi\sqrt{(LC)})$ ✓ $= 1 / (2\pi\sqrt{(0.1 \times 50 \times 10^{-6})})$ ✓ $= 71.18$ Hz ✓ (3)

5.3 Current I drawn along the horizontal reference; V_R in phase with I; V_L 90° ahead of I and V_C 90° behind I, with V_C longer than V_L ✓ (2)
Supply voltage V as the resultant, 38.87° BELOW the current (current leads) ✓

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TOTAL: 100

END OF MARKING GUIDELINE